FLATNESS OF A MODULE IS EQUIVALENT TO THE VANISHING OF THE FIRST TOR FUNCTOR AGAINST EVERY MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: it had been excluded only because the chapter's retained preamble lacks the slogan environment. The statement carries the author's own slogan, which the author's preamble comments out; the card carries a per-card preamble copy formed from the chapter retained preamble plus the project's own three display environments copied verbatim from sources/stacks-chow/preamble.tex lines 7-12.

Editorial note. The counted spans carry no bibliography citation.

Editorial note. The preamble copy described above lives in the card's own task directory and is itself the coordinate of the preamble span.

Editorial note. The label lemma-flat-tor-zero is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Long exact sequence of Tor functors attached to a short exact sequence. Homology, the long exact Tor sequence for modules, used at proof lines 6222-6228. Supplies the identification of the kernel as a quotient of the first Tor group.
- Flat modules are exactly those for which tensoring is exact. Standard, used at proof lines 6220-6221. Supplies the easy direction and the conclusion of the converse.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. The following are equivalent*

(1) *$\mathcal{F}$ is a flat $\mathcal{O}_X$-module, and*
(2) *$Tor_1^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) = 0$ for every $\mathcal{O}_X$-module $\mathcal{G}$.*

Proof. If $\mathcal{F}$ is flat, then $\mathcal{F} \otimes_{\mathcal{O}_X} -$ is an exact functor and the satellites vanish. Conversely assume (2) holds. Then if $\mathcal{G} \to \mathcal{H}$ is injective with cokernel $\mathcal{Q}$, the long exact sequence of Tor shows that the kernel of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \to \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{H}$ is a quotient of $\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{F}, \mathcal{Q})$ which is zero by assumption. Hence $\mathcal{F}$ is flat. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.